TEAM MEMBERS: Ella Goldman, En-Jui (Rachel) Hsu, Michael Morales

Objectives

- Each student must identify their individual objectives and explain how they will contribute to the project.
  - Ella:
    - Administrative tasks:
      - Keeping track of all documents
      - Sharing all documents and making sure all team members are up to speed
      - Keeping a detailed timeline of deliverables
      - Storing tables of progress from each team member
    - Pseudocode, code
    - Debug, test code
    - Complete reflection
    - Presentation
    - Final technical report
    - Simulate code at the end
  - Michael:
    - Github
      - Merging pull requests
      - Pushing to main
      - Organization: readme, etc.
    - Pseudocode, code
    - Debug, test code
    - Complete reflection
    - Presentation
    - Technical report
    - Simulate code at the end
  - Rachel:
    - Research heart rate monitors/data acquisition
    - Pseudocode, code
    - Debug, test code
    - Complete reflection
    - Presentation
    - Final technical report
    - Simulate code at the end

Project objectives:

    - Create a fully-functioning code in C++ that detects given heartbeat values and determines whether they are within a given safe range. If the heart monitor detects that the heart rate is above or below a given threshold, then an alarm will go off and the user will be notified of the next best steps to regulate their heart rate quickly and effectively.
    - Debug the code and confirm that there are no errors. All edge cases need to be accounted for in the final code design.
    - Test different outputs using Linux.
    - Simulate a varying heart rate and confirm that the code is working as intended.
    - Create a written final technical report for the finished code.
    - Design a presentation and present on our code, simulations and findings.

Technologies and Tools

- Programming languages, tools, and frameworks we will use:
    - C++, GitHub, Excel, VS Code, Linux, Google Docs
- Why we decided on these tools:
    - C++: Will be used to code the main part of our project. This is appropriate because it is the programming language that we are most familiar with right now in the course.
    - GitHub: This is where all of our programs will be stored in an organized way. We are familiar with GitHub because we have been using it in the course and have become more comfortable with it in recent projects.
    - Excel: This is where our data for our simulations and tests will be stored, showing how our code could be implemented to effectively test for different heart rates stored in the Excel data.
    - VSCode: All of our C++ code will be written on VSCode. This is the software that we have become comfortable with throughout the course, and it is an effective and user-friendly code editor that we can use to create our final project code.
    - Linux: Linux will be used for our testing and running of our simulated values for our final project. It is an industry-standard software that we have become comfortable with throughout the course and that is readily available for us to use for our project.
    - Google Docs: All administrative tasks and deadlines will be kept track of on Docs, which is easily accessible for all team members and an easy way to keep track of all members and our progress with the project.

Project Timeline

- Identify the major tasks, milestones, and deadlines.

    - Research heart rate monitors, what kind of data inputs
        - Due 10/20
        - Deliverable: Document with list of research findings and sources
    - Create a simulation for heart rate data inputs. Have different ones for different scenarios (workout, resting)
        - Due 10/27
        - Deliverable: Create 3 different data sets for different scenarios.
    - Write code that processes the raw data to find heart beats
        - Due 11/3
        - Deliverable: Functioning cpp code to process raw data uploaded to github
    - Make a functioning code that reads heart rate (finds BPM) and outputs if heart rate is too high/ low
        - Due 11/10
        - Deliverable: Functioning cpp code that takes in input and outputs accordingly uploaded to github
    - Display and log current heart rate, heart rate zones, resting heart rate
        - Due 11/17
        - Deliverable: Decide on display format and show all outputs when run, uploaded to github.
    - Write tests with edge cases
        - Due 11/24
        - Deliverable: All tests showing each different outputs uploaded to github
    - Technical report first draft
        - Due 12/1
        - Deliverable: Rough first draft of report
    - Technical report final
        - Due 12/8
        - Deliverable: Complete final draft of technical report
    - Presentation
        - Due 12/11
        - Deliverable: Complete final presentation
- Assign responsibilities to each team member.
    - Shown above
- Include an expected deliverable for each task.
    - Shown above
- Explain how your team will communicate, track progress, and address challenges.

- Communication primarily through text, call and email, through docs with information on findings

Plan

THIS ITERATION:

Ella	Rachel	Michael
- Divided tasks that needed to be done amongst all team members - Worked on the technologies and tools section - Worked on project objectives section - Documented each team member's contributions on this document - Discussed the dates for the outline	- Divided tasks that needed to be done amongst all team members - Worked on the project timeline section - Discussed the research that needed to be done on heart rate monitors to begin the project - Set due dates for the project timeline while discussing with team	- Divided tasks that needed to be done amongst all team members - Worked on project timeline deliverables in project timeline section - Discussed the proper setup for GitHub and pushing deliverables to the repository - Set up a repository for this iteration.

NEXT ITERATION:

Ella	Rachel	Michael
Keep track of each team member's progress and document contributions of each member in a shared doc, making sure that each member has access to the doc.	Research heart monitors, how they work, and what data they take in and print out.	Create folders within the repository, create rough draft README and CONTRIBUTIONS files.